Global Facilities - Design & Construction Standard -Mechanical - Admin Office Area GHVAC System

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1.0 Purpose

1.1 This document serves as a System Standard document that provides specifications that shall be followed during the Design, Engineering, and Construction of new Administration Area GHVAC Systems as part of new Greenfield FAB Construction projects.

1.2 The standard is applicable for GHVAC Systems provided in Admin/Office Building / Area / Rooms.

1.3 The document is intended to supplement the actual Construction Drawings and Construction Specifications for Construction projects.

2.0 Scope

2.1 This document covers the new Administration Area GHVAC Systems as part of new Expansion Construction projects. This document may or may not apply to existing building space renovation and retrofits of existing Sites, depending on the limitations of the existing Design Concept and System Configuration of those Sites and buildings.

2.2 This document applies to all Micron Facilities employees, contractors, and vendors at all Micron sites.

3.0 Definitions and Acronyms

Term/Acronym	Definition / Description
AHJ	Authority having Jurisdiction
AHU	Air Handling Unit
DOL	Direct On
FCT	Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKMs
FCU	Fan Coil Unit
FMCS/SCADA	Facility Monitoring and Control System / Supervisory Control and Data Acquisition
GFC	Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities
GFTT	Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards
GHVAC	General Heating, Ventilation, Air Conditioning
Global Facilities Central Team	This is used interchangeably with Innovation & Integration Team
Global Facilities Construction Team	This is used interchangeably with Global Construction Team
Global Facilities Operations Support Team	This is used interchangeably with Global Facilities Operations Team
Global Facilities Planning Team	This is used interchangeably with Global Facilities Line Planning Team
LCP	Local Control Panel
LOTO/COHE	Lockout Tagout / Control of Hazardous Energy
MAU	Make up Air Unit
MUA	Make up Air
N+1+F	System sizing concept for equipment quantity consisting of N duty units, +1 redundant standby unit, + future unit(s)
OA	Outside Air
PLC	Programmable Logic Controller
POC/LPOC	Point of Connection / Lateral Point of Connection
POU	Point of Use
UPS	Uninterruptible Power Supply
VAV	Variable Air Volume Device
VSD/VFD	Variable Speed Drive / Variable Frequency Drive

4.0 Roles and Responsibilities

	Responsibilities
GFTT Lead (Discipline Engineer)	• Develop and maintain System Standard document • Ensure Best Known Methods (BKM), lessons learned, and emerging technologies are documented in System Standard • Perform periodic document review and update System Standard
FCT Lead (Discipline Engineer)	• Review and provide technical feedback on System Standard • Support the Site teams by providing knowledge and technical expertise
Site Project Construction Teams	• Review and provide technical feedback on System Standard • Incorporate System Standard into Project Engineering, Design, and Construction • Document project deviations, exceptions, and exclusions from System Standard • Share new lessons learned to GFTT lead, to determine if it is Best Known Method (BKM) that can be captured in future revision of System Standard
Site Facilities Teams	• Use System Standard as technical reference

5.0 Overview of Systems

The following systems are used to support the requirements of the building's ventilation air and heating / cooling loads:

5.1 **Admin/Office Buildings** are buildings / Areas / Rooms that are used for activities that support the operation of company functions and production. Administration rooms are typically used as the offices for daily work routines, meetings, accommodating office asset, etc.

5.2 **Makeup Air (MUA)** is 100% outside air ventilation that is supplied to provide outdoor air ventilation for occupants to meet Indoor Air Quality requirements, and it also provides control of indoor air temperature, humidity, room pressurization level. It compensates for the exhausted air via various exhaust systems or natural exfiltration. The air is conditioned, filtered, and supplied via the Makeup Air Units (MAU).

5.3 **Recirculation Air** is the indoor air that is recirculated through the HVAC system to regulate the room air temperature, humidity, indoor air quality and enhance the air mixing with outdoor makeup air.

5.4 **General HVAC (GHVAC)** System is the air handling system used to control the temperature, humidity, and cleanliness of the air in an enclosed space, in order to provide thermal comfort and acceptable indoor air quality. "HVAC" stands for Heating, Ventilating, Air Conditioning.

5.5 **General HVAC (GHVAC) Exhaust** is used to remove non-corrosive, non-toxic, non-hazardous, or smoke-logged indoor air from the room / area / zone to facilitate air changes rate. This system is a critical part of the room's air conditioning and mechanical ventilation (ACMV) system. This shall not to be confused with General Exhaust (GE) which primarily serves as Production Tools process exhaust in Fab building.

6.0 Performance Specification and System Provision

6.1 The General HVAC (GHVAC) equipment and components shall be as follows:

1. Air Handling Unit (AHU)
2. Makeup Air Unit (MAU)
3. Variable Air Volume (VAV) Devices
4. GHVAC Outdoor Air or Exhaust Fans
5. Air Distribution Ductworks and grilles
6. Air Ductworks Control Dampers

***Note:** Localized Fan Coil Units (FCU) may also be used for remote locations or where deemed necessary, but the preferred basis of design for HVAC configuration shall be Centralized Air Handling Units (AHU).

6.2 GHVAC System design for Admin/Office area shall generally be in line with the industrial recognized standards or guidelines (e.g. ASHRAE, LEED, etc) in terms of the criteria of energy performance and indoor air quality.

6.3 **System Design –** General Heating, Ventilating & Air Conditioning (GHVAC) systems are required to achieve the performance specified in this Standard. While the detail design is on a project-by-project basis, the following concept should be captured:

6.3.1 GHVAC System configuration shall consist of Central Air Handling Units (AHUs), Medium Pressure Supply Air Distribution Ductwork from AHU to VAVs, Variable Air Volume (VAV) Terminal Devices, Low Pressure Supply Air Distribution Ductwork, and Ceiling

Diffuser Terminal Devices. Return Air shall be through Ceiling Return Grilles and above-ceiling Return Plenum.

6.3.2 MAU Ventilation Airflow supply shall be designed accordingly to facilitate the required ventilation rate of the occupied areas. MAU Ventilation Airflow can be supplied directly to the central AHU units, where the Ventilation Airflow will mix with the Return Airflow. This mixed air will then be distributed as supply air to all of the occupied areas.

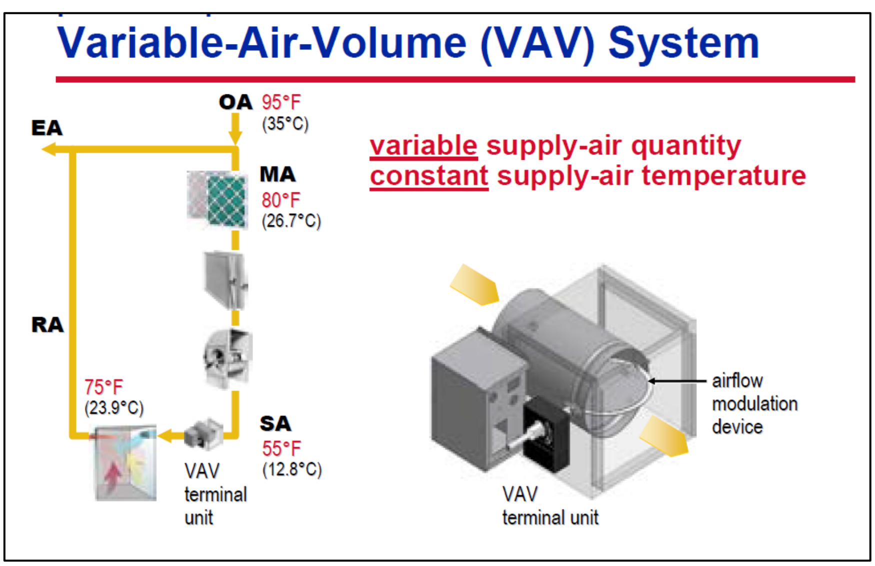
6.3.3 AHU / MAU Systems shall be designed with ductworks & fittings (e.g. VAV boxes, motorized dampers, sensing elements, control valves, etc) to have proper provision to ventilate, pressurize and regulate individual room air condition.

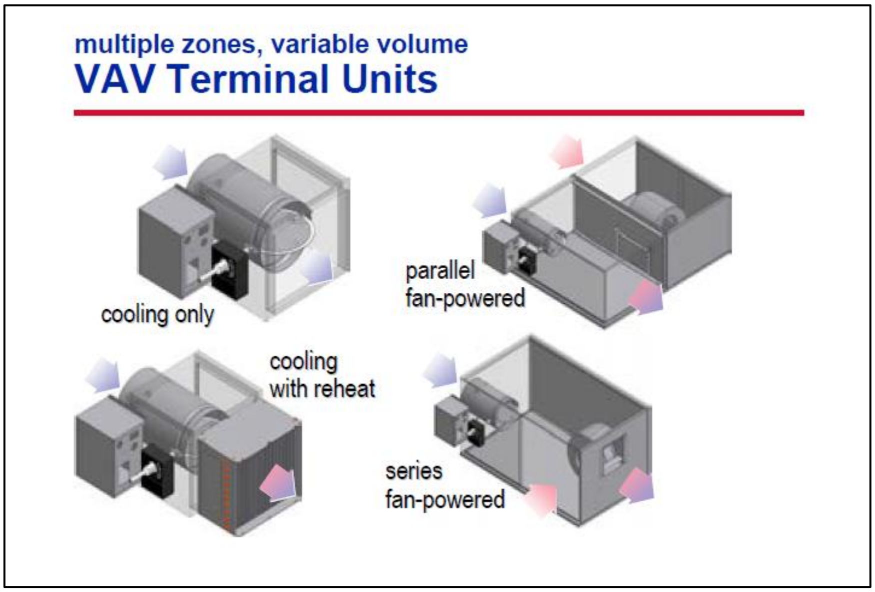
6.3.4 Fan motors should be designed with VSD to regulate its fan speed and airflow to operate at designated output corresponding to the required ventilation rate of each rooms (or zones).

6.3.5 AHU / MAU Systems design concept shall also be aligned with the smoke extraction strategy (i.e. de-smoke system) during the fire events or emergency cases.

6.3.6 Supply Air Distribution Ductwork completed with Variable Air Volume (VAV) Devices shall be designed with automatic supply air volume modulation to ensure the air flowrate (and/or air pressure) can properly be balanced to achieve the design flowrate, based on the room or area cooling load. The VAV Devices can be cooling only type or with reheat type, depending on the room or area cooling & heating load requirements (i.e., interior vs perimeter spaces).

Example GHVAC Configuration with Central AHU and VAV Boxes (Courtesy of Trane)





The Admin/Office Area GHVAC Systems shall be designed and installed to achieve the required minimum room (i.e. breathing zones) ventilation rate, room condition and etc as outlined below and in accordance with ASHRAE requirements; See **Table 1**, **Table 2 and Table 3.**

Table 1: Breathing Zone Outdoor Airflow

$$V_{bz} = R_p \times P_z + R_a \times A_z$$

Where,

R_p = outdoor airflow rate required per person, as in Table 2
P_z = zone population, largest expected number of occupants during typical usage
R_a = outdoor airflow rate required per unit area, as in Table 2
A_z = zone floor area, m2

Table 2: Minimum Ventilation Rate in Breathing Zone / Rooms

Rooms / Areas	People outdoor Air Rate, *Rp* (L/s·person)	Area outdoor Air Rate, *Ra* (L/s·m2)	Minimum Exhaust Rates
Corridors	-	0.3 (1.1 cmh/m^2)	-

Offices	2.5 (9.0 cmh/person)	0.3 (1.1 cmh/m^2)	-
Meeting Rooms / Conference Rooms	2.5 (9.0 cmh/person)	0.3 (1.1 cmh/m^2)	-
Main Entry Lobbies	2.5 (9.0 cmh/person)	0.3 (1.1 cmh/m^2)	-
Cafeteria / Dining	3.8 (13.7 cmh/person)	0.9 (3.2 cmh/m^2)	-
Multipurpose Room / Assembly Hall	2.5 (9.0 cmh/person)	0.3 (1.1 cmh/m^2)	-
Kitchen (Cafeteria)	-	-	3.5 (12.6 cmh/m^2)
Pantry (Kitchenette)	-	-	1.5 (5.4 cmh/m^2)
Toilet	-	-	25 (L/s·unit) (90.0 cmh/unit) ***Note 3**

Note 1: The GHVAC system shall be designed and installed with capability to meet the required minimum ventilation rates stated above, or the mandatory ventilation rates by Local AHJ, whichever is higher or more stringent.

Note 2: The minimum ventilation rates listed above are referencing to ASHRAE 62.1.

Note 3: The rate is per unit of water closet, urinal, or both.

Note 4: The GHVAC system shall be designed and installed in accordance with **FCT Guideline – Future Office HVAC & In-duct UVC Guideline.**

Table 3: Design Criteria of Room Condition

Rooms / Areas	Room Condition (Temperature, Relative Humidity)		Remarks
	°C	RH%	
Corridors	20 ~ 26	No control	
Main Entry Lobbies	20 ~ 26	≤65%	
Offices	20 ~ 26	≤65%	
Meeting Rooms / Conference Rooms	20 ~ 26	≤65%	
Break Rooms	20 ~ 26	≤65%	

Cafeteria	20 ~ 26	≤65%	
Mechanical Equipment Rooms	24 ~ 28	No control	Refer to Global Standard of Utility Room GHVAC System

Note 1: The HVAC System shall be designed and installed with capability to meet the required room condition stated above, or in compliance with those required as recommended by Local AHJ, whichever is more stringent. The local climate profile shall also be taken into consideration to design an energy efficient system as recommended (or mandatory) by AHJ.

Note 2: Engineering calculation to be provided to substantiate the design cooling load assigned to each rooms / areas.

Table 4: Special requirement per COVID-19 Indoor Air Quality recommendations

Area / System	Description
Bi-Polar Ionization or Ultra-Violet Lighting Disinfection Systems	• Bi-Polar Ionization (BPI) or Ultra-Violet (UV) Lighting disinfection system shall be installed in all GHVAC system serving occupied areas to eliminate air-borne bio-hazardous microorganisms, viruses, and pollutants to improve the indoor air quality. • BPI and/or UV disinfection systems to be considered for rooms / areas where the Occupants will be in close proximity (e.g., administrative office, lobby, gym, cafeteria, office areas in Store / Warehouse / Shipping, etc.) • For rooms / areas where there are also Instrument & Equipment which are sensitive to ionized air molecules (e.g., Lab with office / Probe / Production (Non-Cleanroom area), BPI system shall not be used, and only the UV disinfection system shall be used.

Note: The special requirement is specified as per the COVID-19 Indoor Air Quality recommendations. It shall be in accordance with **FCT Guideline – Future Office HVAC & In-duct UVC Guideline.**

6.4 Performance Specification of GHVAC AHU / MAU

6.4.1 While the design of GHVAC AHU / MAU is highly dependent on the local weather profile and room load requirements, the design parameters listed in **Table 5** shall serve as the guideline value to maintain the design consistency across Micron Admin Buildings.

Table 5: Performance Specification for GHVAC AHU / MAU

Parameter	Specification Requirement	Remarks
Supply Air Temperature	16 ~ 23°C (60.8 ~ 73.5 °F)	• Measured at outlet of MAU / AHU • Subject to local atmospheric condition

Note: The AHU/MAU System shall be designed and installed with capability to meet the required performance specification stated above. Actual operational parameters should be considered on a case-by-case basis as per the local atmospheric condition and room load requirements, as well as the actual ducting design.

7.0 Equipment Sizing & Design Requirements:

7.1 **Redundancy** – GHVAC System main equipment, namely AHU / MAU / FCU and Fans shall be designed:

7.1.1 Since GHVAC Systems are serving non-critical non-Production spaces, providing redundancy and N+1 backup provisions is not a requirement.

7.1.2 For the AHU / MAU system serving a room or a group of rooms (or zones), the units shall be in the arrangement of N per the design capacity (with N being single unit or multiple units).

7.1.3 The design redundancy shall be in compliance with the requirement of AHJ if it is more stringent than above items 7.1.1 and 7.1.2.

7.2 **System Diversity and Safety Factor –** GHVAC System sizing shall be selected so that the N capacity meets 100% of the system demand and space load requirements at full build (after initial construction of the system), but also applying a safety factor of 15-20%.

7.2.1 To determine the system demand at N full build, the peak demand load shall be based on the Building's HVAC load calculation. The estimation of HVAC load shall be considering, fenestration, infiltration, internal heat generation by occupants and equipment, the local weather profile, and etc.

7.2.2 Make-up air shall be considered accordingly to provide ventilation for occupants for indoor air quality requirements and in accordance with ASHRAE, compensate for the exhausted air, to maintain required room condition, as well as to regulate the room air pressure.

7.2.3 See **Table 6** for AHU Cooling / Heating Load criteria and calculation**, Table 7** for GHVAC Outdoor Air / Exhaust system sizing criteria and calculation, and **Table 8** for AHU system sizing criteria and calculation.

Table 6: System Sizing Calculation – AHU System

AHU is the main HVAC component to treat the indoor air to maintain the Room Condition at the specified thermal comfort zone.

- Cooling Load Demand at Full Build, D_C, shall consider the following components of heat gain:

  D_C = Heat gain through building envelopes + heat gain through fenestration + heat gain through outdoor air infiltration + heat load generated (sensible + latent) by Occupancy

- Heating Load Demand at Full Build, D_H, shall consider the following components of heat loss:

  D_H = Heat Loss through building envelopes + heat loss through indoor air exfiltration

- AHU System N_C cooling capacity:
  $N = D_C \times 1.2$ (Safety Factor)

- AHU System N_H heating capacity:
  $N = D_H \times 1.2$ (Safety Factor)

Table 7: System Sizing Calculation – GHVAC Outdoor air / Exhaust System

<u>With reference to Table 1</u>

- GHVAC Outdoor air / Exhaust System Demand at Full Build, D:
  D = Total Fresh air / Exhaust Flow Rate

- GHVAC Outdoor air / Exhaust System N capacity:
  $N = D \times 1.2$ (Safety Factor)

Table 8: System Sizing Calculation – MAU System

With reference to Table 1 • MAU System Demand at Full Build, D: D = Total Required Outdoor Air Flow Rate **Actual MUA supply flowrate should take into consideration of the design air pressure level (i.e. relatively positive or negative) of each room / compartment. Air pressure level is typically maintained by having a different flowrate between make-up air and exhaust air. • MAU System N capacity: $N = D_C \times 1.2$ (Safety Factor) **Actual MUA cooling / heating load calculation shall take into consideration of the outdoor air condition of the local weather profile.

7.3 **Equipment Quantity –** GHVAC system main equipment (per room / area) shall be selected for N units per the design capacity (with each unit capacity = 1/N of 100%, N≥2). Each unit to be equal in capacity sizing. Thus, a common GHVAC system serving a group of rooms / areas will be advantageous as it is more cost and space effective by reducing the total unit quantity.

7.4 **Equipment Layout –** Equipment shall be suitably located to comply with all equipment maintenance clearances and access requirements. Adequate space and a clear routing path shall be reserved for future move-in of new equipment, and to allow move-in and move-out of large components during maintenance and repair.

7.4.1 Outdoor air / Exhaust Fans shall have adequate clearance space to allow for maintenance, repairs, and/or replacement on fan assembly, impeller, and motor.

7.4.2 AHU / MAU shall have adequate clearance space to allow for maintenance, repairs, and/or replacement on filters, coils, fan assembly, impeller, and motor.

7.4.3 Local FCU / VAV boxes shall be located in accessible spaces above ceiling or on floor (for FCU), and shall have adequate clearance space to allow for maintenance, repairs, and/or replacement on filters, coils, fan assembly, impeller and motor.

7.4.4 Equipment should be connected to a common header / plenum in such a way that minimizes the pressure drop, and to optimize the redundancy of the entire system.

7.5 **Equipment Location -** The location of the AHU / MAU / Ventilation Fans should be designed as such to reduce the length and pressure drop of the ductworks.

7.5.1 HVAC equipment shall be located away from Office areas or areas for occupants' circulation to minimize interrupting the daily operation in these areas during the maintenance of the equipment.

7.5.2 HVAC equipment shall be located and installed as such to minimize the operating noise propagation into Office areas.

7.5.3 The HVAC equipment should be accessible for the routine inspection and maintenance. The HVAC equipment is recommended to be in an indoor equipment room. The ductworks should be planned to avoid passing other rooms (or zones) to minimize the risk of spreading of hazards (e.g. fire / smoke).

7.6 **Pressure Drop Calculations –** Submit pressure drop calculations for each ductwork system, to show that each fan selection and the fan parameters (flow, pressure, power, etc.) are adequate to meet the design requirements.

7.7 **Equipment Selection Criteria** - AHU / MAU / FCU system selection criteria and requirements are as follow:

7.7.1 AHU / FCU shall consist of some of the following coils listed below, dependent on design location climate, and load profile. The associated Coil water source shall be as follows:

1. Primary cooling coil: MTCHW

7.7.2 MAU shall consist of some of the following coils listed below, dependent on design location, climate, and load profile. The associated Coil water source shall be as follows:

1. Pre-heating coil (where required): Warm water
2. Primary cooling coil: MTCHW
3. Secondary cooling coil (where required): LTCHW
4. Re-heating coil (where required): Warm water

7.7.3 All coil sections shall have stainless steel sloped drain pans. From the cooling coil section on the interior shall be out of stainless steel. Face velocity through the unit approx. 2.5 m/s, through heat exchanger ≤ 2.5 m/s.

7.7.4 The units will have one access side for maintenance, repair and exchange of components. All components will be accessible via access doors with windows. The doors will have airtight gasket seals and have latches or hinges.

7.7.5 Each AHU or MAU shall have minimum 2 filter sections:

- Pre-filter: rated at 30 to 35% (MERV 5 or G3)
  Recommend pressure drop: 65/130 Pa (clean/dirty)
- Secondary filter: rated at 80 to 85% (MERV 13 or F7)
  Recommend pressure drop: 150/250 Pa (clean/dirty)

7.7.6 Fan motor sizing and selection shall be as follows:

- Fans shall be Direct Drive where possible, instead of Belt Drive.
- Fan motors shall be selected for non-overloading operation on all points of the performance curve at the design conditions.
- Fan motors shall be rated for inverter duty with variable speed operation.
- Fan motor efficiency rating shall be minimum IE3.
- Fan motor IP rating shall be minimum IP55. If located outdoors, fan motor IP rating shall be IP56.
- Fan shall have flexible ductwork connection on inlet side and flexible ductwork connection on discharge side.

8.0 Duct Design and Sizing Requirements:

8.1 **Ductwork Diversity and Sizing Factor** – The ductwork shall be sized according to the same sizing factors stated in the above Section 7.2 and below.

8.2 **Ductwork Mains Sizing Calculation** – Duct Mains shall be sized to accommodate of the airflow demand at full build (after initial construction of the system). See **Table 9**.

Table 9: Duct Mains Sizing Calculation

Ductwork to be sized to have its static pressure drop across all branches to be almost equal • GHVAC Airflow Demand at Full Build, D: D = Total Airflow Rate **

** Total airflow rate may include normal operating airflow and/or emergency ventilation airflow (e.g., make-up airflow for smoke extraction system, if applicable)

- GHVAC Duct Mains sizing:
  Duct sizing = N+1+F; to include Future units

8.3 **Duct Sizing Criteria –** Main Duct Sizing shall be selected based on the full build capacity, including the future equipment (N+1+F). It is recommended for the duct sizing to be designed with maximum specific pressure drop not to exceed the following:

8.3.1 Low Pressure Supply Air Duct – 0.6 Pa/m

8.3.2 Medium Pressure Supply Air Duct – 0.8~1.0 Pa/m

8.4 **Duct Pressure Rating, –** Ductwork mains shall be designed to meet the minimum pressure class in accordance with SMACNA standards, such as +3 inches w.g. (+750 Pa). If the system pressure at the main ductwork exceeds this value, then the ductwork pressure rating shall meet or exceed the system pressure.

8.5 **Duct Supply Diffusers and Return / Exhaust Air Grilles** – The Air Diffusers and Grilles should evenly be distributed to have an effective air distribution & recirculation within the room.

8.5.1 Supply Diffusers and Return Air Grilles should be positioned at the top part of the room.

8.5.2 Supply Diffusers and Return Air Grilles should be strategically positioned to enhance mixing of the conditioned air with the room air. Position of these grilles should avoid air path short-circuit.

8.5.3 Return / Exhaust Air Grilles should generally be positioned at the top part of the room and to avoid prematurely exhausting the supply air from the Supply Air Grille.

8.6 **Duct Outdoor Air intake / Exhaust Air Outlet** – Outdoor air intakes shall not be within 5m from any exhaust discharge openings / outlets.

8.6.1 The distance from an air intake to a cooling tower is measured from the base of the cooling tower.

8.6.2 Outdoor air intakes shall not be within 10m from any exhaust discharge openings / outlets of noxious or potentially hazardous exhaust.

8.7 **Materials of Construction** - GHVAC Ductwork, rectangular and circular, shall be made of Galvanized Steel, in accordance with SMACNA requirements. Ductwork to be insulated from the equipment unit outlet connection up to the supply air grille in the rooms. The ductwork insulation material shall be vapor-tight, with a closed cell structure, made of expanded cellular polyethylene foam

- **Gaskets:** Neoprene tape, silicon-free gaskets shall be used for GHVAC ductwork.

8.8 **Ductwork Installation** - Low pressure rectangular ductwork shall be made of galvanized steel sheet, in accordance with SMACNA requirements. The ductwork system shall be complete with transitions, bends, tees, supports, dampers, offsets, flexible connections, take-offs and similar fittings necessary for the balancing and full operation of the air distribution system. Ductwork shall be installed in accordance with SMACNA "Industrial Duct Construction Standards."

- Flexible duct connection length shall be limited to 5 feet (1.5m) and without sharp elbows or bends

8.9 **Fire Dampers and Fire-rated enclosure**

8.9.1 Fire Dampers shall be installed in ductworks wherever there is penetration of fire compartment.

8.9.2 If Fire Dampers cannot be installed in ductworks due to its nature of operation (e.g., kitchen exhaust duct), fire-rated enclosure on the ductworks should be considered as the alternative provision to maintain the fire compartment integrity.

Note: If the local code or jurisdiction prohibits the installation fire dampers in the ductwork, it shall take precedence.

9.0 **Other Accessories -** Provide other accessories as outlined in **Table 10**.

Table 10: Accessories to be installed

Accessories	Type & Installation Location
Insulation	• Ductworks shall be insulated wherever there is potential occurrence of air condensation (internal or external of ductworks) • Install for SA ductworks • Install for all CHW piping • Install for condensate drain piping

Accessories	Type & Installation Location
Ductwork Supports	• Provide duct hanger and support material gauges in compliance with details on the Drawings. Where details are not provided, comply with SMACNA Industrial Duct Construction Standards. • Ductwork hanger and suspension material may be threaded rod, structural shapes, or strut material. The use of wire as suspension material is prohibited.
Piping Supports	• Install piping hangers, supports, racks, and clamps at the appropriate spacing interval as required by the system and piping load and layout • Provide seismic restraints and supports as deemed required during design, based on site location and local jurisdiction requirements • Provide vibration isolation for piping supports as deemed required during design, based on the building vibration criteria
Pressure Gauges	• Install at cooler supply and return connection for CHW • Install at Heater supply and return connection for WW
Temperature Gauges	• Install at cooler supply and return connection for CHW • Install at Heater supply and return connection for WW
Pressure Difference Gauges	• Install at each filter section
Air Vents	• Install at cooler supply and return connection for CHW • Install at Heater supply and return connection for WW
Drain Ports	• Install at all low point locations
Valves	• All butterfly valves shall be rated for “bi- directional” duty with positive shutoff in both directions • All butterfly valves shall be lug-style
Control Valves	• Provide control valve type and quantity as outlined in this standard
Labeling and Identification	• Provide identification labels at minimum of every 6m distance and change of direction

Accessories	Type & Installation Location
	• Refer to the specific project's Construction Specification, (i.e. Spec Section 15190 or equivalent Spec) for additional requirements. **<u>Ductwork</u>** • "SA" nomenclature shall be used for air-conditioning Supply Air • "RA" nomenclature shall be used for air-conditioning Return Air • "MAU" nomenclature shall be used for air-conditioning Make-Up air • "GHVAC Exhaust" nomenclature shall be used for General HVAC Exhaust

10.0 Sustainability Design:

10.1 **Ductwork Pressure Drop Sizing –** It is recommended to increase the ductwork sizes where possible, to minimize the air velocity and pressure drop through the ductwork. This will result in lower fan energy consumption.

10.2 **VAV devices** regulate the airflow to each rooms / zones based on the actual loading and this can conserve energy by reducing fan and refrigeration power.

10.3 **LEED Design –** GHVAC System design in line with LEED guideline recommendation will enhance the overall system sustainability in terms of energy efficiency and indoor environmental quality (IEQ).

11.0 Electrical Design:

11.1 Electrical power supply system for all GHVAC equipment shall be designed and installed in compliance with **Global Facilities - Design & Construction Standard - Electrical - Electrical Design Specification**.

11.2 **Power Segregation –** Electrical power supply for GHVAC equipment shall be segregated among a minimum of 2 separate upstream power sources (wherever possible). The quantity of equipment served from each upstream power source shall be split evenly

11.3 **GHVAC system and system controls** used for Admin/Office Area comfort cooling and ventilation shall not require Emergency Power or Un-interrupted Power (UPS).

11.4 **GHVAC system and system controls** used for Life Safety Systems and Emergency Systems shall be connected to Emergency Power or Un-interrupted Power (UPS), as required by Global Standards or Local Code Regulations.

11.5 **Variable Frequency Drives –** All central AHU and MAU fans shall be VFD driven for variable speed operation. Variable Frequency Drives shall be selected to be one size larger than the power rating of the actual fan motor size. Local FCU fans may not require VFDs.

11.5.1 Variable Frequency Drives shall be compliant with SEMI standard F47 for voltage sag immunity, to provide ride-through capability during momentary power interruption or voltage sag.

11.5.2 Variable Frequency Drives shall be configured to “Fail Last Speed” configuration upon any failure, including but not limited to, loss of power, loss of communications, etc.

11.6 **Motors -** Fan motors shall be inverter duty rated with insulated bearings and shaft grounding kits. Motors shall be compatible with the associated VFD.

11.6.1 All motors located outdoors shall be IP56 rated protection.

11.6.2 All motors shall be National Electrical Manufacturers Association (NEMA) IE3 rated or better.

11.7 **Local Disconnect Switches –** Each fan as part of the AHU / MAU / FCU unit shall have a lockable type, local disconnect switch installed adjacent to the fan, to allow for manual isolation and lockout/tagout (LOTO/COHE) of the fan. All disconnect switches must be reasonably accessible for the authorized personnel to safely isolate the power supply to facilitate maintenance works.

12.0 Instrumentation:

12.1 The following Instrumentation shall be installed and monitored/controlled via FMCS: See **Table 11**.

Table 11 - Instrumentation for AHU / MAU

Instrumentation	**Type and Installation Location**	
	AHU	**MAU**
Pressure Transmitters	• Install at AHU Common Main Supply header (Quantity: 1 duty)	• Install at Main header (Quantity: 1 duty)

Differential Pressure gauges	• Install at each filter bank (Quantity: 1 duty)	• Install at each filter bank (Quantity: 1 duty)
Temperature / Humidity Transmitters	• Install at inlet and outlet of AHU, (Quantity: 2 duty Temperature transmitter, 1 duty Humidity Transmitter)	• Install at outlet of MAU (Quantity: 1 duty Temperature transmitter, 1 duty Humidity transmitter) • Can be converted to Dewpoint & Enthalpy
Temperature Transmitter (in unit)	• Nil	• Install at outlet of heating and Cooling coils (Quantity: 1 duty each)
CO_2 Transmitter	• Install at inlet of AHU, (Quantity: 1 duty CO_2 Transmitter)	• Nil
Room HVAC (Temperature / /Humidity Transmitter)	• Room shall have air temperature and humidity sensors for VAV operation and SCADA monitoring and control purposes	

12.2 Instrumentation points of connection shall be provided with a means of isolation (ball valve, etc.) for maintenance and/or replacement.

12.3 Smoke detectors / Instruments that are monitoring for smoke shall be located as such that the detection signal capable of identifying the specific zone of the incident (e.g. in the Return Air Duct). AHUs serving the specific zone of the incident shall be stopped if the smoke detector is triggered to prevent the smoke spreading into other areas.

12.4 The Pressure Differential Transmitters on ductwork shall reference the Zero Reference Pipe (ZRP). This shall be the same ZRP used for Make-up air unit pressurization reference.

12.4.1 ZRP shall reference outdoor atmospheric pressure. ZRP external probe shall be located in a place with minimum wind interference and shall have adequate cover protection from wind.

13.0 Control Sequence Philosophy:

13.1 The overall philosophy for system control and operation shall be listed in this section. See **Table 12**.

Table 12: System Control Logic

Control Parameter	Control Logic
GHVAC Ventilation Fans Start/Stop	• Ventilation Fans Start/Stop shall be done automatically via SCADA or hardwired logic, based on required number of running fans • Ventilation Fans Start/Stop/Rev shall be capable of being remotely controlled via SCADA by Operation Personnel.
AHU / FCU	• AHU / FCU Start/Stop shall be done automatically via SCADA based on required number of running units • **VSD Speed** of AHU Fans shall be modulated to maintain the pressure setpoint at the Main duct. VSD speed of FCU Fans (if applicable) shall be modulated based on the room load requirements. **Cooling** • Position output of 1st cooling coil temperature control valve shall be modulated to maintain the temperature setpoint at Return Air Duct / Supply Air Main Header.
MAU	• MAU Start/Stop shall be done automatically via SCADA based on required number of running units • **VSD Speed** of MAU Fans shall be modulated to maintain the pressure setpoint at the Main duct. **Preheating** • Preheating coil (if necessary) temperature control valve shall be modulated to pre-heat the outdoor air to the temperature set point at outlet of preheating coil. **Humidifier** • Humidifier (if necessary) control valve shall be modulated to the humidity set point at the outlet of the humidifier.

Control Parameter	Control Logic
	Cooling / Dehumidification • Position output of 1st cooling coil temperature control valve shall be modulated to maintain the temperature (and dewpoint) setpoint at outlet of 1st cooling coil. **Reheating** • Position output of reheat coil (if necessary) control valves shall be modulated to maintain the relative humidity (RH%) setpoint at outlet of MAU.
Ductwork Motorized Dampers	• Motorized dampers opening percentage to be measured and confirmed through an air-balancing activity. • Motorized dampers opening percentage to be actuated to rated percentage, either during Normal or Emergency mode, to allowed for proper air flowrate. • Normal / Emergency mode rated flowrate to be validated by manual measurement.

13.2 A detailed Sequence of Operation (SOO) shall be developed specifically for each system, as part of each construction project.

13.3 Control Valves and Dampers

13.3.1 The following control valve and damper fail positions shall be configured. See **Table 13**.

Table 13: GHVAC System Motorized Dampers and Control Valves

Control Valve or Damper	Control Valve or Damper Fail Position		
	Loss of Air	**Loss of Electricity**	**Loss of Control Signal**
AHU / MAU Ductworks Motorized Dampers	Fail Last State	Fail Last State	Fail Last State
AHU / MAU Temperature Control Valves	Fail Last State	Fail Last State	Fail Last State

13.3.2 The control valves and dampers shall be double-acting type actuators, capable of failing to last state position and holding the position for unlimited duration without needing control air, power, or PLC signal.

13.3.3 After a loss event of control air, power, or PLC signal, and after it is restored, the control valve or damper output command/position shall be set to the same command/position it was operating at prior to the loss event.

13.3.4 All control valves and dampers shall have a manual override bypass, to allow manual operation of the valve or damper upon a failure.

13.3.5 For motorized dampers with actuators, it is recommended to have square-shaped shafts connecting the damper and the actuator, to prevent slippage and mis-operation. Prior to system startup, perform inspection and operations check of all dampers and actuators including shaft, linkages, and connection points.

14.0 Inter-Discipline Coordination Matrix:

14.1 The following provisions shall be provided by each of the following disciplines and coordinated accordingly: See **Table 14**.

Table 14: Inter-Discipline Coordination Matrix

Discipline	Provision
Architectural	• Provide Mechanical Room space for GHVAC Equipment. • All rooms shall be constructed with appropriate fire compartmentation and fire rated partition / wall. • Cut openings in walls and floors for ducting penetrations, and seal openings after duct installation • Provide adequate fire stopping at all required fire wall penetrations • Provide sloped floors and floor drains in Mechanical Room space containing the GHVAC equipment.
Mechanical	• Provide GHVAC space conditioning for Mechanical Room space containing the GHVAC equipment. Room temperature shall be designed to maintain the space temperature range of 24°C to 28°C. • Provide heating water, chilled water and city water supply piping, and accessories as outlined in Standards, to provide sufficient heating, cooling water for GHVAC equipment

Discipline	Provision
	• Provide drain water piping, floor drains and accessories to be located suitably to allow proper draining of equipment, condensate drains and auto- drains, piping, and air vents • Louver shall be constructed to blend in Architectural façade design, e.g. color, structure, etc. • Exhaust louver shall be constructed at least 5 meter away from any outdoor air intake / building opening. • Outdoor air louver shall be constructed at least 5 meter away from any exhaust louver. • Fire dampers shall be installed in ductworks wherever it penetrates the fire compartment.
Electrical	• Provide local disconnect switches for all fan motors. • Provide power wiring for all Electrical equipment/ components • Provide grounding for all Electrical equipment/ components • Provide Emergency Power / UPS Power as outlined in this Standard • Provide Power Segregation as outlined in this Standard • Provide VFDs for all fan motors. • Provide lightning protection for outdoor equipment including exhaust discharge stacks.
Controls	• Provide control damper and control valve actuators and positioners • Provide control wiring • Provide instrument control air tubing • Provide Zero Reference Pipe (ZRP) for atmospheric pressure reference of Exhaust and Makeup Air Systems • Provide FMCS/SCADA system monitoring and controls of all instrumentation and transmitters including pressure, temperature, flow, gas detector, fire detector etc. • Provide FMCS/SCADA system connectivity of all Local Control Panels

15.0 Quality Inspection and Testing:

15.1 **Inspection and Test Plan (ITP) –** Create formal Inspection and Test Plans (ITP) that outline all requirements for quality verification, inspections, testing, and commissioning:

15.1.1 Piping and Ductwork

- Piping and Ductwork Shop Drawings and Submittals
- Incoming Material Inspection
- Material Storage
- Assembly Procedures, Qualifications, and Inspection
- Duct Supports Installation and Verification
- QA/QC Inspection & Punchlist
- Ductwork Leak Testing –At least 20% HVAC ductwork shall be leak tested.
- Leak Testing and Pressure Testing
- Condensate piping installation shall be maintained with proper gradient to ensure the gravity flow is sufficient to drain out the condensate.
- Quality Certification
- Final QA/QC Inspection & Punchlist
- Training, Documentation, and Handover

15.1.2 Equipment

- Equipment Shop Drawings and Submittals
- Factory Acceptance Test
- Incoming Equipment Inspection
- Equipment Move-In and Installation
- QA/QC Inspection & Punchlist
- Equipment Pre-Startup Checks – Mechanical, Electrical, Controls, etc.
- Operational Acceptance Test (OAT) - Standalone Equipment Startup per Vendor’s Procedure to allow equipment to run in MANUAL mode. Also considered “Early Run” Equipment.

15.1.3 System

- Site Acceptance Test (SAT) – Test System operation and control functions in accordance with P&ID, SOO, and Specifications.
- Integrated System Test (IST) – Test full System operation including all system interdependencies, interlocks, and critical failure modes.
- Final QA/QC Inspection & Punchlist
- Training, Documentation, and Handover

16.0 Startup Commissioning / Site Acceptance Test:

16.1 At a minimum, the following system control functions shall be tested and successfully proven during the System Startup Testing & Commissioning / Acceptance Tests.

16.1.1 Fans (including for AHU/MAU/FCU/VAV/Exhaust & Ventilation Fans)

- Fan Motor Rotation Check
- Fan/Motor Alignment Check and Lubrication
- All Damper, Valve, and Actuator Checks
- Motorized Damper operation checks including actuator, shaft, linkages, and connection points
- VSD Settings and Functionality Checks
- Control interface and communication checks to FMCS/SCADA
- Fan Standalone Startup Procedure per Manufacturer's procedure
- Fan Vibration Level Checks
- Verify Fans Operation through its operating range (0 to 50/60 Hz)
- Verify Fan Start/Stop Function to Rotate Operating Fan
- Verify Fan Auto-Start sequencing, upon failure of an operating Fan
- Verify VSD fails last speed upon any failure
- Verify control interface and communication checks to FMCS/SCADA
- Control interface and communication checks to Local Control Panel, PLC, and FMCS/SCADA.

16.1.2 AHU / MAU / FCU /VAV

- Confirm the external static pressure and the flowrate
- Confirm system pressure control is stable during fan manual start/stop rotation, and fan auto-start procedure
- Confirm Auto-switch to backup pressure transmitter(s) upon failure of main pressure transmitter(s)
- Confirm all control valves and dampers fail to the stated position per this Standard, upon loss of instrument air, power, and/or control signal.
- Verify all dampers and coil control valves operation through its operating range (0-100%)
- Confirm all unit temperature sensors and freezestat operation
- Confirm all UV light and Bi-Polar Ionization function and operation
- Confirm all smoke detectors operation and interlock functions
- Confirm space temperature, humidity, and CO2 sensors monitoring and operation
- Air flow test and balancing for all rooms, VAV boxes, duct branches, SA diffusers & RA grilles and exhaust grilles.
- Verify all alarms are enabled and activated in SCADA

- Verify equipment on emergency power is able to operate on emergency power, after simulated loss of normal power
- Perform Comprehensive Power loss and recovery (blackout) test

16.1.3 Integrated system performance test

- Confirm system pressure control is stable during fan manual start/stop rotation, and fan auto-start procedure
- Confirm Auto-switch to backup pressure transmitter(s) upon failure of main pressure transmitter(s)
- Confirm all control valves and dampers fail to the stated position per this Standard, upon loss of instrument air, power, and/or control signal.
- Verify all alarms are enabled and activated in SCADA
- Verify equipment on emergency power is able to operate on emergency power, after simulated loss of normal power
- Perform Comprehensive Power loss and recovery (blackout) test
- system integrated testing, AHU / MAU / Exhaust System and control / interlock with motorized dampers activation.

17.0 Reference Specifications:

17.1 The following Construction Specifications shall be referenced during design, engineering, and construction. The Specifications shown below are in accordance with the Construction Specifications Institute (CSI) Master Format; Refer to the specific Construction Specifications for each project. See **Table 17**.

Table 17 – Specification Sections

Spec Section	Spec Contents
Section 15060	Pipe and Pipe Fittings
Section 15190	PVC and CPVC Piping Systems
Section 15101	Valves
Section 15120	Piping Specialties
Section 15121	Process piping Specialties
Section 15130	Indicating Devices
Section 15140	Supports and Anchors
Section 15190	Mechanical Identification
Section 15240	Mechanical Sound and Vibration Control
Section 15260	Piping Insulation
Section 15280	Equipment Insulation
Section 15330	Wet Pipe Sprinkler Systems
Section 15461	Process Exhaust System
Section 15515	Hydronic Specialties
Section 15540	Pumps
Section 15860	Exhaust Fans
Section 15861	FRP Exhaust Fans and Stacks
Section 15891	Metal HVAC Ductwork and Accessories
Section 15892	Metal Industrial Ductwork and Accessories
Section 15993	Air Systems Testing, Adjusting, and Balancing
Section 15170	Motors
Section 15171	Variable Frequency Drives

Document Control

17.2 Approval: GLOBAL_FAC_GFTT_APPR

17.3 Notification:

GLOBAL_FAC_MANAGERS
GLOBAL_FAC_COMMUNICATIONS
GLOBAL_FAC_CONSTRUCTION
GLOBAL_FAC_NOTIFY
FCG_F4_FAC_NOTIFY
FCG_F6_FAC_NOTIFY
FCG_F10_FAC_NOTIFY
FCG_F11_FAC_NOTIFY
FCG_F15_FAC_NOTIFY
FCG_F16_FAC_NOTIFY
FCG_MMY_FAC_NOTIFY
FCG_MMP_FAC_NOTIFY
FCG_MSB_FAC_NOTIFY
FCG_MTB_FAC_NOTIFY
FCG_MXA_FAC_NOTIFY

17.4 Final Viewer Access

All Micron Team Members at all Sites

17.5 Retention

Adherence to the requirements specified in
Global Facilities - Document Control Procedures

17.6 Review

Biennially (two years)

20.0 Revision History

Revision #	Section Changed	Details of Changes	Date
0	NA	New document	05/05/2022